

Evolutionary Game Theory (EGT)

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Definition

Evolutionary Game Theory (EGT) is a branch of game theory that studies how strategies evolve over time within a population. Instead of assuming that players are perfectly rational and always make the best decisions, **EGT assumes that successful strategies become more common because they lead to better outcomes (higher fitness or payoff).**

Originally developed by biologists to explain animal behavior, it is now widely used in economics, political science, psychology, computer science, artificial intelligence, and business.

Key Idea

In classical game theory, players **think before acting**.

In evolutionary game theory, players **follow strategies**, and strategies that perform better **spread over time**, similar to natural selection.

Simple Formula:

Better strategy → Higher payoff → More likely to be copied → Becomes more

Core Concepts

1. Population

Instead of two individuals making decisions, there is an entire population.

Example:

- 1,000 animals
- 10,000 consumers
- Millions of social media users

Each individual follows one strategy.

2. Strategy

A strategy is a fixed behavior.

Examples:

- Cooperate
- Cheat
- Fight
- Share
- Stay loyal
- Defect

3. Fitness (Payoff)

Fitness measures how successful a strategy is.

Higher payoff means:

- More resources
- More profit
- Higher survival
- Greater influence

Strategies with higher fitness become more common.

4. Evolution

Individuals usually do not change strategies through careful reasoning.

Instead:

- Successful strategies are copied.
- Unsuccessful strategies gradually disappear.

❖ Evolutionarily Stable Strategy (ESS)

The most important concept in Evolutionary Game Theory.

Definition

An **Evolutionarily Stable Strategy (ESS)** is a strategy that, once adopted by most members of a population, cannot be successfully invaded by a small group using a different strategy.

In simple words:

If almost everyone follows Strategy A, then Strategy B cannot outperform it and replace it.

Conditions for ESS

Strategy **S** is an ESS if:

1. No alternative strategy earns a higher payoff against S.
2. If another strategy performs equally well, S performs better against the alternative than the alternative performs against itself.

Examples

Example: Business Competition

Two companies sell similar products.

Aggressive Pricing

- Heavy discounts
- Price wars

Cooperative Pricing

- Maintain stable prices

If every company starts price wars:

- Profits fall.
- Everyone suffers.

Over time, businesses may settle into a stable pricing pattern that balances competition and profitability.

Example: Politics

Political parties choose between:

- Cooperation
- Polarization

If extreme polarization causes voters to lose confidence, parties that cooperate and solve problems may gain more support. If voters reward aggressive campaigning, confrontational strategies may become more prevalent.

Classical Game Theory

Assumes rational decision-makers

Players consciously choose actions

Focuses on Nash Equilibrium

Often analyzes one-time interactions

Common in economics

Evolutionary Game Theory

Assumes strategies evolve over time

Strategies spread through success

Focuses on Evolutionarily Stable Strategy (ESS)

Often studies repeated evolution in populations

Common in biology, AI, economics, sociology, and political science

Applications

Evolutionary Game Theory is widely applied in:

- **Biology:** Animal behavior, evolution, cooperation.
- **Economics:** Market competition, pricing strategies.
- **Artificial Intelligence:** Multi-agent systems and reinforcement learning.
- **Political Science:** Voting behavior and coalition formation.
- **Cybersecurity:** Attack–defense strategy evolution.
- **Public Health:** Vaccination decisions and disease control.
- **Environmental Science:** Managing common resources and encouraging sustainable behavior.

Advantages

- Explains behavior without assuming perfect rationality.
- Models how behaviors change over time.
- Useful for studying large populations.
- Applicable across many disciplines.
- Captures long-term dynamics rather than one-time choices.

Limitations

- Assumes strategy frequencies change mainly through success or imitation.
- Real human decisions can also be influenced by emotions, ethics, and incomplete information.
- Predicting long-term evolution can be mathematically complex.
- Real populations may not always reach a stable equilibrium.

